

SGMS Workspace Grid Modelling Microservice API Contract

API Reference Documentation

Version 1.0.0

January 4, 2026

1. Introduction

The SGMS Workspace Grid Modelling Microservice provides comprehensive workspace management and electrical grid modelling capabilities. This API enables users to create, manage, and analyze electrical grids within collaborative workspaces.

1.1 Base URL

All endpoints are relative to the base URL of the workspace grid modelling service.

1.2 Authentication

The API uses cookies for authentication. All endpoints (except the welcome endpoint) require a cookie header with its value:

```
cookie: <access_token>
```

2. API Endpoints

2.1 Welcome Endpoint

- **Endpoint:** GET /
- **Description:** Welcome message endpoint.
- **Response:** 200 OK with welcome message

2.2 Workspace Management

2.2.1 List Workspaces

- **Endpoint:** GET /api/v1/workspaces/
- **Description:** Retrieve all workspaces for the authenticated user
- **Response:** 200 OK with list of workspaces

2.2.2 Create Workspace

- **Endpoint:** POST /api/v1/workspaces/
- **Description:** Create a new workspace
- **Request Body:**

```
{
  "name": "string",
  "description": "string (optional)",
  "visibility": "private|shared"
}
```

- **Responses:**
 - **201 Created:** Workspace created successfully
 - **422 Validation Error:** Invalid input data

2.2.3 Get Workspace

- **Endpoint:** GET /api/v1/workspaces/{workspace_id}
- **Description:** Retrieve workspace details with associated grids
- **Parameters:**
 - workspace_id (UUID): Workspace identifier
- **Response:** 200 OK with workspace details

2.2.4 Update Workspace

- **Endpoint:** PUT /api/v1/workspaces/{workspace_id}
- **Description:** Update workspace properties
- **Parameters:**
 - workspace_id (UUID): Workspace identifier
- **Request Body:**

```
{
  "name": "string (optional)",
  "description": "string (optional)",
  "visibility": "private|shared (optional)"
}
```

- **Response:** 200 OK with updated workspace

2.2.5 Delete Workspace

- **Endpoint:** DELETE /api/v1/workspaces/{workspace_id}
- **Description:** Delete a workspace
- **Parameters:**
 - workspace_id (UUID): Workspace identifier
- **Response:** 200 OK

2.3 Grid Management

2.3.1 Create Grid

- **Endpoint:** POST /api/v1/grids/workspaces/{workspace_id}/grids
- **Description:** Create a new electrical grid within a workspace
- **Parameters:**
 - workspace_id (UUID): Workspace identifier

- **Request Body:**

```
{
  "name": "string",
  "description": "string (optional)",
  "base_power_mva": "number",
  "voltage_level": "number",
  "is_active": "boolean"
}
```

- **Responses:**
 - **201 Created:** Grid created successfully
 - **422 Validation Error:** Invalid input data

2.3.2 List Grids for Workspace

- **Endpoint:** GET /api/v1/grids/workspaces/{workspace_id}/grids
- **Description:** Retrieve all grids within a workspace
- **Parameters:**
 - workspace_id (UUID): Workspace identifier
- **Response:** 200 OK with list of grids

2.3.3 Get Grid

- **Endpoint:** GET /api/v1/grids/{grid_id}
- **Description:** Retrieve grid details with nodes and edges
- **Parameters:**
 - grid_id (UUID): Grid identifier
- **Response:** 200 OK with grid details and relationships

2.3.4 Update Grid

- **Endpoint:** PUT /api/v1/grids/{grid_id}
- **Description:** Update grid properties
- **Parameters:**
 - grid_id (UUID): Grid identifier
- **Request Body:**

```
{
  "name": "string (optional)",
  "description": "string (optional)",
  "base_power_mva": "number (optional)",
  "voltage_level": "number (optional)",
  "is_active": "boolean (optional)"
}
```

- **Response:** 200 OK with updated grid

2.3.5 Delete Grid

- **Endpoint:** DELETE /api/v1/grids/{grid_id}
- **Description:** Delete a grid
- **Parameters:**
 - grid_id (UUID): Grid identifier
- **Response:** 200 OK

2.4 Grid Nodes Management

2.4.1 Create Grid Node

- **Endpoint:** POST /api/v1/grids/{grid_id}/nodes/
- **Description:** Add a node to a grid (plant, load, or substation)
- **Parameters:**
 - grid_id (UUID): Grid identifier

- **Request Body:**

```
{
  "name": "string",
  "type": "plant|load|substation",
  "plant_type": "hydro|solar|thermal (optional)",
  "load_type": "hospital|enterprise|residential|industrial (optional)",
  "demand_mw": "number (optional)",
  "capacity_mw": "number (optional)",
  "priority": "integer (optional)",
  "x": "number (optional)",
  "y": "number (optional)"
}
```

- **Response:** 201 Created with node details

2.4.2 Get Grid Node

- **Endpoint:** GET /api/v1/grids/{grid_id}/nodes/{node_id}
- **Description:** Retrieve node details with connected edges
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - node_id (UUID): Node identifier
- **Response:** 200 OK with node details and connections

2.4.3 Update Grid Node

- **Endpoint:** PUT /api/v1/grids/{grid_id}/nodes/{node_id}
- **Description:** Update node properties
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - node_id (UUID): Node identifier
- **Request Body:**

```
{
  "name": "string (optional)",
  "plant_type": "hydro|solar|thermal (optional)",
  "load_type": "hospital|enterprise|residential|industrial (optional)",
  "demand_mw": "number (optional)",
  "capacity_mw": "number (optional)",
  "priority": "integer (optional)"
}
```

- **Response:** 200 OK with updated node

2.4.4 Update Grid Node Position

- **Endpoint:** PATCH /api/v1/grids/{grid_id}/nodes/{node_id}/position
- **Description:** Update node position coordinates
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - node_id (UUID): Node identifier
- **Request Body:**

```
{
  "x": "number",
  "y": "number"
}
```

- **Response:** 200 OK

2.4.5 Delete Grid Node

- **Endpoint:** DELETE /api/v1/grids/{grid_id}/nodes/{node_id}
- **Description:** Remove a node from a grid
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - node_id (UUID): Node identifier
- **Response:** 200 OK

2.5 Grid Edges Management

2.5.1 Create Grid Edge

- **Endpoint:** POST /api/v1/grids/{grid_id}/edges/
- **Description:** Create a connection (transmission line) between nodes
- **Parameters:**
 - grid_id (UUID): Grid identifier
- **Request Body:**

```
{
  "from_node_id": "UUID",
  "to_node_id": "UUID",
  "capacity_mw": "number",
  "resistance": "number (optional)",
  "losses_percent": "number (optional)",
  "priority": "integer",
  "status": "active|disabled|failed"
}
```

- **Response:** 201 Created with edge details

2.5.2 Get Grid Edge

- **Endpoint:** GET /api/v1/grids/{grid_id}/edges/{edge_id}
- **Description:** Retrieve edge details
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - edge_id (UUID): Edge identifier
- **Response:** 200 OK with edge details

2.5.3 Update Grid Edge

- **Endpoint:** PUT /api/v1/grids/{grid_id}/edges/{edge_id}
- **Description:** Update edge properties
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - edge_id (UUID): Edge identifier
- **Request Body:**

```
{
  "capacity_mw": "number (optional)",
  "resistance": "number (optional)",
  "losses_percent": "number (optional)",
  "priority": "integer (optional)",
  "status": "active|disabled|failed (optional)"
}
```

- **Response:** 200 OK with updated edge

2.5.4 Update Grid Edge State

- **Endpoint:** PATCH /api/v1/grids/{grid_id}/edges/{edge_id}/state
- **Description:** Update edge operational status
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - edge_id (UUID): Edge identifier
- **Request Body:**

```
{
  "status": "active|disabled|failed"
}
```

- **Response:** 200 OK with updated edge

2.5.5 Delete Grid Edge

- **Endpoint:** DELETE /api/v1/grids/{grid_id}/edges/{edge_id}
- **Description:** Remove a connection between nodes
- **Parameters:**
 - grid_id (UUID): Grid identifier
 - edge_id (UUID): Edge identifier
- **Response:** 200 OK

2.6 Grid Import

- **Endpoint:** POST /api/v1/import/grids/{workspace_id}
- **Description:** Import grid configuration from file
- **Parameters:**
 - workspace_id (UUID): Workspace identifier
- **Request Body:** multipart/form-data with grid file
- **Response:** 201 Created

3. Data Models

3.1 Workspace Models

3.1.1 WorkspaceCreate

```
{
  "name": "string",
  "description": "string (optional)",
  "visibility": "private|shared"
}
```

3.1.2 WorkspaceUpdate

```
{
  "name": "string (optional)",
  "description": "string (optional)",
  "visibility": "private|shared (optional)"
}
```


3.1.3 WorkspaceOut

```
{
  "id": "UUID",
  "name": "string",
  "description": "string (optional)",
  "owner_id": "UUID",
  "visibility": "private|shared",
  "created_at": "datetime",
  "updated_at": "datetime (optional)"
}
```

3.2 Grid Models

3.2.1 GridCreate

```
{
  "name": "string",
  "description": "string (optional)",
  "base_power_mva": "number",
  "voltage_level": "number",
  "is_active": "boolean"
}
```

3.2.2 GridOut

```
{
  "id": "UUID",
  "name": "string",
  "description": "string (optional)",
  "base_power_mva": "number",
  "voltage_level": "number",
  "is_active": "boolean",
  "workspace_id": "UUID",
  "created_at": "datetime",
  "updated_at": "datetime (optional)"
}
```

3.3 Grid Node Models

3.3.1 GridNodeCreate

```
{
  "name": "string",
  "type": "plant|load|substation",
  "plant_type": "hydro|solar|thermal (optional)",
  "load_type": "hospital|enterprise|residential|industrial (optional)",
  "demand_mw": "number (optional)",
  "capacity_mw": "number (optional)",
  "priority": "integer (optional)",
  "x": "number (optional)",
  "y": "number (optional)"
}
```

3.3.2 GridNodeOut

```
{
  "id": "UUID",
  "name": "string",
  "type": "plant|load|substation",
  "plant_type": "hydro|solar|thermal (optional)",
  "load_type": "hospital|enterprise|residential|industrial (optional)",
  "demand_mw": "number (optional)",
  "capacity_mw": "number (optional)",
  "priority": "integer (optional)",
  "x": "number (optional)",
  "y": "number (optional)",
  "grid_id": "UUID",
  "created_at": "datetime",
  "updated_at": "datetime (optional)"
}
```

3.4 Grid Edge Models

3.4.1 GridEdgeCreate

```
{
  "from_node_id": "UUID",
  "to_node_id": "UUID",
  "capacity_mw": "number",
  "resistance": "number (optional)",
  "losses_percent": "number (optional)",
  "priority": "integer",
  "status": "active|disabled|failed"
}
```

3.4.2 GridEdgeOut

```
{
  "id": "UUID",
  "from_node_id": "UUID",
  "to_node_id": "UUID",
  "capacity_mw": "number",
  "resistance": "number (optional)",
  "losses_percent": "number (optional)",
  "priority": "integer",
  "status": "active|disabled|failed",
  "grid_id": "UUID",
  "created_at": "datetime",
  "updated_at": "datetime (optional)"
}
```

4. Enumerations

4.1 WorkspaceVisibility

- **private:** Only owner can access
- **shared:** Accessible to multiple users

4.2 GridNodeType

- **plant:** Power generation facility
- **load:** Power consumption facility
- **substation:** Power distribution/transformation facility

4.3 PlantType

- **hydro:** Hydropower plant
- **solar:** Solar power plant
- **thermal:** Thermal power plant

4.4 LoadType

- **hospital:** Hospital facility load
- **enterprise:** Business/office load
- **residential:** Residential area load
- **industrial:** Industrial facility load

4.5 GridEdgeStatus

- **active:** Operational transmission line
- **disabled:** Deactivated/out-of-service line
- **failed:** Failed/outage condition

5. Error Handling

5.1 HTTPValidationError

```
{
  "detail": [
    {
      "loc": ["string" or "integer"],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

5.2 Common HTTP Status Codes

- **200 OK:** Request successful
- **201 Created:** Resource created successfully
- **400 Bad Request:** Invalid request parameters
- **401 Unauthorized:** Authentication required
- **403 Forbidden:** Insufficient permissions
- **404 Not Found:** Resource not found
- **422 Unprocessable Entity:** Validation error
- **500 Internal Server Error:** Server error

6. Security Considerations

- All endpoints require JWT authentication (except welcome)
- Workspace visibility controls access (private/shared)
- Users can only access workspaces they own or are shared with them
- Grid modifications are logged for audit purposes
- File imports are validated for security

7. Rate Limiting

API requests are subject to rate limiting to prevent abuse. The current limits are:

- Workspace operations: 10 requests per minute per user
- Grid operations: 20 requests per minute per user
- Node/Edge operations: 30 requests per minute per user operations: 5 requests per minute per user

8. Implementation Notes

- All IDs are UUIDv4 format
- Grids must have unique names within a workspace
- Nodes require type-specific fields (plant_type for plants, load_type for loads)
- Edges require valid from/to node references within the same grid
- Grid import supports common electrical grid file formats (PSS/E, CIM, etc.)
- Position coordinates (x, y) are optional for visualization
- Priority values determine operation/load shedding order
- Base power (MVA) and voltage level (kV) are required for grid calculations